

BORONG XU

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Research Assistant at SLAI

ABOUT ME

- Research Assistant at the Shenzhen Loop Area Institute (SLAI) and a top 3% graduate of HKUST's Computer Engineering (CPEG) program. Passionate about robotics and embodied intelligence, with research experience spanning visual-inertial odometry, SLAM, multi-camera localization, and quadruped control, as well as deep learning, vision transformers, and LLM-based agent systems. Eager to push the boundaries of autonomous systems in real-world deployments.

EDUCATION

The Hong Kong University of Science and Technology (HKUST)

Hong Kong SAR

BACHELOR OF ENGINEERING IN COMPUTER ENGINEERING

Graduated: June 2026

- GPA 3.84/4.3 (Top 3% among 157 students)
- Major GPA 4.03/4.3

High School Affiliated to Renmin University of China (RDFZ)

Beijing, China

HIGH SCHOOL DIPLOMA(GAOKAO)

Graduated June 2022

- Activities:
 - FIRST Robotics Club, Chief Engineer (season 2021)
 - FIRST Robotics Club, Alumni Mentor (season 2022-2024)

EXPERIENCE

Shenzhen Loop Area Institute (SLAI)

Shenzhen, China

RESEARCH ASSISTANT

Jun 2026 - Present

- Research on robotics and embodied intelligence.

PROJECT EXPERIENCE

LLM Agent Development

Supervised by Prof. Shenghui Song

UNDERGRADUATE RESEARCH OPPORTUNITY (UROP)

Jun 2024 - May 2026

- Designed and implemented a Large Language Model (LLM) agent capable of playing board games. Conducted comprehensive evaluations of agent strategies and performance.

ELEC5660 Introduction to Aerial Robotics

Supervised by Prof. Shaojie Shen

COURSE PROJECT

Feb 2026 - May 2026

- Developed aerial robot algorithms from scratch, including visual-inertial odometry, Extended Kalman Filter (EKF) state estimation, and autonomous path planning within a simulation environment, enabling robust UAV localization, mapping, and navigation.

Quadruped Robot Control

Supervised by Prof. Maurice Pagnucco and Prof. Yang Song

RESEARCH PRACTICUM

Feb 2025 - Aug 2025

- Built and programmed a Unitree GO2 quadruped robot to perform complex tasks, integrating depth camera for accurate object localization (localization error < 10cm) and LiDAR for SLAM.

Underwater Localization System

Supervised by Prof. Huan Yin and Prof. Fumin Zhang

FINAL YEAR PROJECT

Jun 2025 - May 2026

- Developed a Multi-Camera Vision-Based Localization System for Tracking Multiple Autonomous Underwater Vehicles (AUVs). Reduced localization error in complex underwater lighting scenarios, improving positioning accuracy to within 10cm in a 1m x 1m range.

HONORS

2022-2026 Dean's List, 5 times

HKUST

2023-2026 Continuing Undergraduate Students Scholarship, 10,000 HKD per year

HKUST

2023 Bronze Medal, The Innovation Competition: Our Livable City, We Engineer

HKIE

2022 Entry Scholarship, 70,000 HKD

HKUST



SKILLS

- C/C++:** Eigen, PCL, STL, STM32, ESP32
- Python:** PyTorch, OpenCV, Numpy
- ROS/ROS2:** Navigation2, Gazebo, Slam Toolbox, TF
- Robotics:** SLAM, VIO, EKF, ICP, Path Planning, Control
- Deep Learning:** YOLO, ViT, RNN, GAN, VAE, Diffusion, LLM
- Machine Learning:** GLM, SVM, PCA, XGBoost, HMM
- Hardware:** Circuit Design, Soldering, CAD, 3D Printing

LANGUAGES

English • Fluent (IELTS 7.5)
Mandarin • Mother tongue